How Drill Bits Work

01 Which part makes the hole?



The cutting lips at the tip shave off chips.

Look closer. Look at the cutting edges at the tip.

A little explanation. A twist drill bit is a metal rod with spiral grooves. As it turns and moves into the material, the cutting lips at its tip shave off tiny bits called chips.

Try this. Trace the tip in the picture, then follow one spiral groove.

02 Where do the chips go?



Flutes carry chips up and out of the hole.

Look closer. Follow the chips from the bottom of the hole to the top.

A little explanation. The spiral grooves are called flutes. They help carry chips up and out as the bit turns, keeping the hole clear. The cutting lips make chips; the flutes give them a way out.

Try this. Use the diorama to follow one chip up the flute.

03 What can we notice by turning slowly?



A screw thread helps a turning screw move into the soap.

Look closer. Look for a curl beside the screw thread.

A little explanation. A screw has a spiral ridge called a thread. It acts like a ramp wrapped around the rod. Turning against the material helps the screw move in. A screw thread is a raised ridge; a drill flute is a groove.

Try this. With Big close by, hand-turn a screw slowly into soap or clay on a tray and look at the curls. Power drills are tools for adults; kids do not use them. Big stays close. Try turning a pencil in an enclosed hand sharpener, too.

04 What else can a turning spiral move?



An Archimedes screw lifts water to a higher place.

Look closer. Water rides from a low bowl to a higher cup.

A little explanation. An auger lifts loose soil or ice from a hole. An Archimedes screw lifts pockets of water along a sloping tube. A spiral slopes along its length, so turning against material can push or carry that material along. It needs something to work against, not just empty space.

Try this. Wrap a ribbon around a pencil and follow its sloping path with your eyes.